

CSE477 — Data Mining Lab 3 Manual

Title: Pattern Discovery from Cleaned YouTube Text (Frequent Word Co-occurrence & Contextual Insights)

Objective: This lab introduces students to the idea of frequent word pattern discovery using cleaned YouTube comment/caption data. It blends manual inspection, algorithmic mining, and reflection. The goal is not to teach Apriori or pattern mining theory formally, but to enable meaningful exploration of token co-occurrence from the datasets processed in Lab 2.

Estimated Task Count: 57

A. Setup & Data Recall

Tutorial:

What is co-occurrence?

When two or more words often show up together across multiple rows of data (like many people mentioning "exam" and "stress" in different comments), we say those words co-occur. It's a way to catch repeated connections without reading everything manually.

Tasks:

1. Load `cleaned_comments.csv` and `cleaned_captions.csv` into separate pandas DataFrames.
2. Verify that the `cleaned_tokens` column exists and contains list-like data.
3. Drop rows where `cleaned_tokens` is missing or empty.
4. Use `collections.Counter` to print the top 20 unigrams.
5. Install `mlxtend` using `pip` if not already installed.

Helpful tools:

- Google: "co-occurrence meaning in text mining simple explanation"
- LLM Prompt: "Explain co-occurrence in text analysis using a simple YouTube example."

B. Transaction Construction

Tutorial:

What is a transaction in text mining?

A transaction is like a shopping basket — just a bunch of things together. If a YouTube comment says: "love this channel, great work", then its transaction might be `[love, channel, great, work]`. Order doesn't matter.

Tasks:

6. Treat each list in `cleaned_tokens` as one basket (transaction).
7. Remove baskets with fewer than 3 tokens.
8. Optionally remove duplicate tokens within each basket.
9. Assign a unique transaction ID to each basket.
10. Store the cleaned transactions as a list of lists.
11. Plot a histogram of basket lengths.
12. Print statistics: average, min, max basket length.

Helpful tools:

- Google: "what is a transaction in market basket analysis"
- LLM Prompt: "Convert a list of tokenized comments into transactions for Apriori algorithm."

C. Manual Pattern Discovery

Tutorial:

Co-occurrence vs. frequency

Word frequency tells us how often a word appears. Co-occurrence tells us how often two words appear together in different comments. For example, "change" might be common, but seeing "climate" and "change" often together is a co-occurrence pattern.

Tasks:

13. Iterate over all baskets to count pairwise word co-occurrences.
14. Use a dictionary or `Counter` to store pair counts.
15. Keep only those pairs with count ≥ 3 .
16. Plot the top 20 co-occurring word pairs as a horizontal bar chart.
17. Compare with a unigram frequency bar chart.
18. Create a network graph with `networkx` (words as nodes, edges = co-occurrence).
19. Apply further filtering to remove stopwords from co-occurring pairs.
20. Export the co-occurrence data to CSV for reuse.

Helpful tools:

- Google: "python count word co-occurrence in list of lists"
- LLM Prompt: "How to visualize co-occurrence of words using matplotlib or networkx?"

D. Algorithmic Pattern Discovery

Tutorial:

What is Apriori?

Apriori is a method that finds groups of words that often appear together. If many users

mention [earth, climate, change] together, Apriori detects that. Then it tries to form rules like: *If a comment has 'climate' and 'change', it probably also includes 'earth'.*

Tasks:

21. Convert your transaction list to a one-hot encoded DataFrame.
22. Run Apriori with `min_support = 0.3` and extract itemsets.
23. Repeat with `min_support = 0.2` and `0.1`.
24. Filter itemsets by length (keep 2 and 3 item sets).
25. Run `association_rules()` from `mlxtend` to calculate confidence and lift.
26. Filter rules with confidence ≥ 0.6 .
27. Filter rules with lift ≥ 1.2 .
28. Plot support vs confidence scatter plot.
29. Save frequent itemsets to `frequent_itemsets.csv`.
30. Save association rules to `association_rules.csv`.

Helpful tools:

- Google: "mlxtend apriori one-hot example"
- LLM Prompt: "How to generate and filter association rules with mlxtend?"

E. Exploration Using Variations

Tutorial:

Why small changes matter

If you tweak one number — like minimum support — suddenly you get many more patterns, or almost none. This shows how sensitive these algorithms are. It also teaches you to experiment, not just follow blindly.

Tasks:

31. Try `min_support = 0.15` and observe new patterns.
32. Try `min_support = 0.05` and check if too many trivial patterns appear.
33. Swap out lemmatization for stemming.
34. Remove tokens under 4 characters — then re-run Apriori.
35. Apply the entire pipeline on `cleaned_captions.csv`.
36. Merge comments + captions and rerun Apriori.
37. Compare rules from merged vs separate datasets.
38. Reflect: Which patterns vanished or appeared under new settings?

Helpful tools:

- LLM Prompt: "What factors affect frequent itemsets in Apriori algorithm?"

F. Pattern Interpretation & Visualization

Tutorial:

Interpreting what we find

Finding patterns is not enough — we need to understand them. For instance, if “online” and “lecture” appear together a lot, that might signal students commenting about online classes. Interpretation turns data into insight.

Tasks:

39. Plot bar chart of top 10 2-itemsets by support.
40. Plot top 10 3-itemsets by confidence.
41. Create a word cloud from most frequent items.
42. Build a simple cluster graph of word associations.
43. Compare patterns from captions vs comments.
44. Write three insight statements based on observed rules.
45. Label each insight with support/confidence metrics.

Helpful tools:

- Google: "interpreting association rules examples"
- LLM Prompt: "Give examples of how to interpret association rules in simple language."

G. Wrap-Up & Reflection

Tutorial:

Why reflection matters

Pattern mining is useful only if we take time to think: Which patterns were helpful? Which were noise? What do we do next? Reflection is where insight begins.

Tasks:

46. Save final cleaned dataset with itemsets and rules.
47. Save all plots as PNG images.
48. Answer reflection Q1: “What’s a surprising pair you found?”
49. Answer reflection Q2: “What was hard about finding patterns?”
50. Answer reflection Q3: “What would you explore next?”
51. Compare your answers with a sample response.
52. Submit all files in a folder named `Lab3_PatternMining_YourName`.
53. Prepare for next lab: retain cleaned data and rules file for reuse.

Helpful tools:

- LLM Prompt: "What are good questions to reflect on after data mining patterns?"